

aws

COMMUNITY DAY

AHMEDABAD

28th February 2026



Turning S3 Buckets into Queryable Insights: A Rapid Data Lake Walkthrough

Ashwin Raiyani
AWS Community Builder



 @ashwinraiyan



Today's Journey

- 1 The Problem - Data Swamps & Why They Happen
- 2 The Solution - Serverless Services
- 3 Demo - Building a Governed Data Lake



The Real Problem – Data Sitting Idle in S3

AWS S3

Buckets

my-corporate-data-lake

raw-data

raw-data

purchases

Warning: Public access is blocked by bucket-level settings. However, some objects might be public.

Amazon S3 > purchases/

Objects (1)

Objects warning: Public access is blocked by bucket-level settings. However, some objects might be public.

Search purchases/

	Name	Last modified	Size	Storage class
☒	purchase_data_final_v2_new_20231026.csv	2023-10-26	9 KB	Poory
☐	logs_output.txt	2023-10-26	9 KB	Poory
☐	purchase_records_test.json	2023-10-26, 10:15:52 UNT	0 KB	Poory
☐	raw_purchases_dump.csv	2023-10-26		
☐	temp_file.tmp	2023-10-26		
☐	purchase_data_final.csv	2023-10-26		

Actions

ix

lit

Actions

Query with S3 Select

Query with Athena

Permissions: Default "read/write" settings apply to this folder. No granular governance or access controls are defined.

Query Attempt Failed

ERROR: Cannot query object 'purchase_data_final.csv'. No schema definition or AWS Glue Table found for s3://my-corporate-data-lake/raw-data/purchases/. Please define a schema and catalog your data before querying.

Formats

Complee S3

Public access current bucket-level settings.

Privacy

Tiles

Nanorn statement

Rotesucks

Citkeresh

© 2022. S3. console.aws.amz201e33anon.

Tones

Privacy anil situations



Can't query it without downloading files!

5TB

Slow download

Where is the customer_data_v2_final_final.csv?
No searchable catalog!

Log

S3

DATA SWAMP

Credit Card
Email
SSN

Unsecure open

Everyone sees PII!

No Access Control!

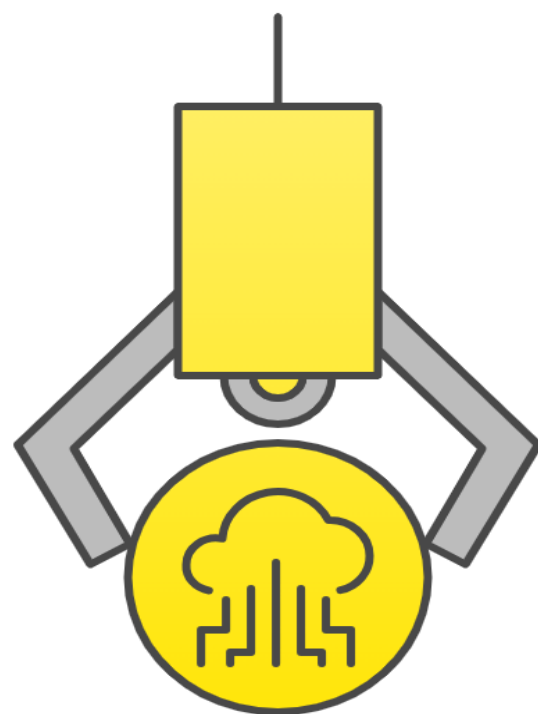
Analysts wait DAYS for simple answers!
⌚ Wait... Wait... Wait...

Result: Data Swamp, not Data Lake



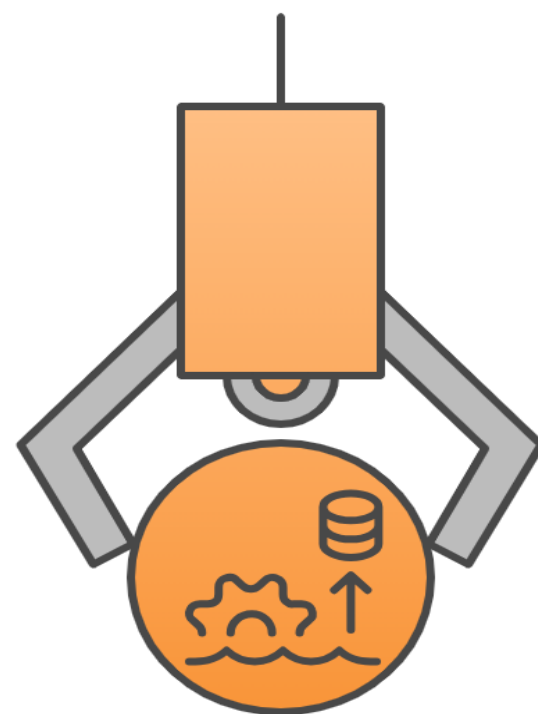
The ultimate solutions

Serverless Data Lake Components



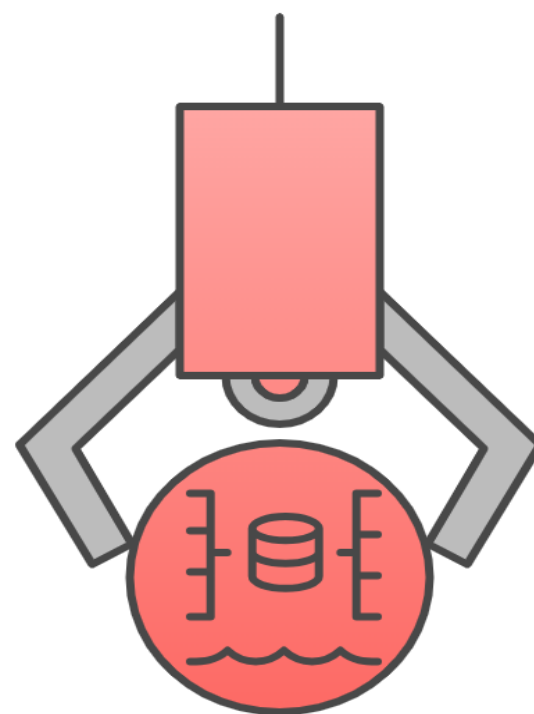
Amazon S3

Storage foundation
for the data lake.



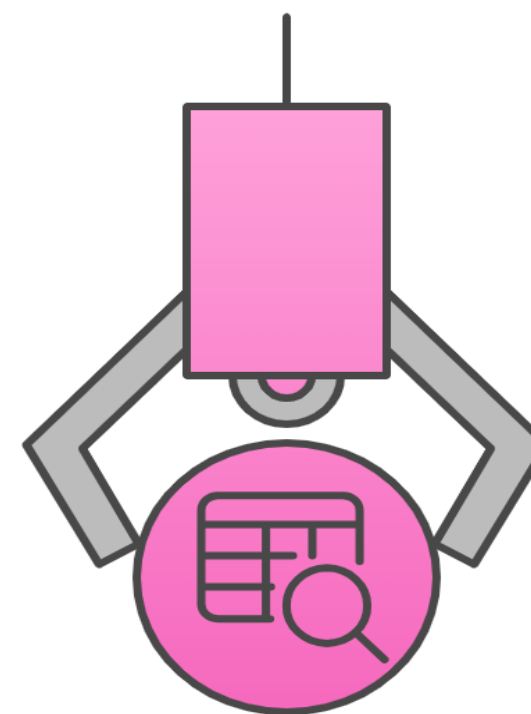
AWS Glue

Schema discovery
and cataloging of
data.



Lake Formation

Fine-grained access
control for data
security.



Amazon Athena

Serverless SQL
queries for data
analysis.



AWS
User Groups
Ahmedabad

Cloud-integrated Storage

Bucket: datalake-demo-acd-26/
 └─ customer_data/
 └─  customer_orders.csv

#2. AWS Glue - Automatic Schema Discovery

What Glue Does:



1. Crawls your S3 data



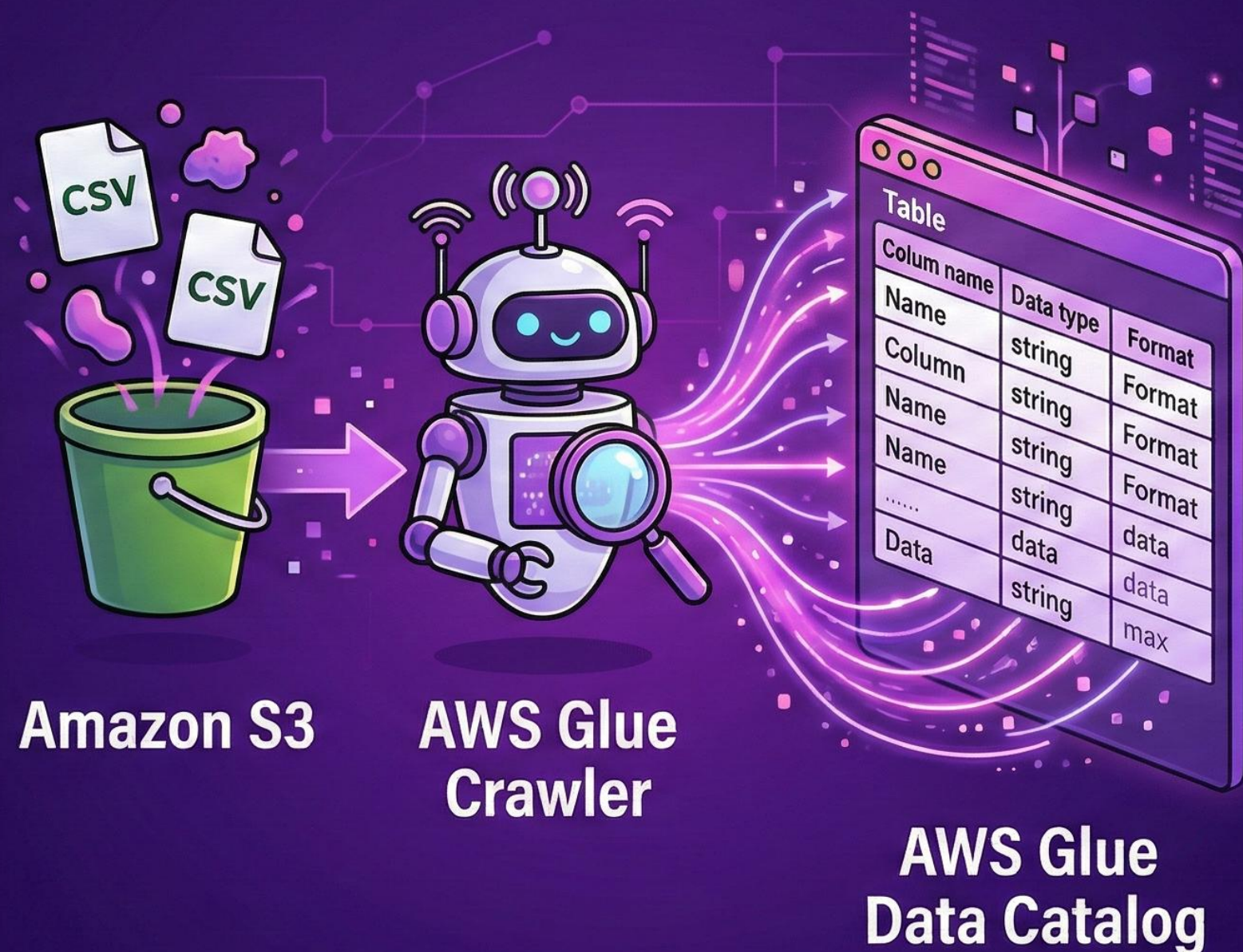
2. Infers schema automatically



3. Creates searchable catalog



4. No manual DDL!



Glue Crawler:



- Points at S3 path



- Detects: column names, types, formats



- Creates: queryable tables



- Runs: on-demand or scheduled

Database Creation

Create a database

Create a database in the AWS Glue Data Catalog.

Database details

Name

Database name is required, in lowercase characters, and no longer than 255 characters.

Description - optional

Descriptions can be up to 2048 characters long.

Database settings

Location - optional

Set the URI location for use by clients of the Data Catalog.

An S3 location is required for managed tables and Zero-ETL integrations.

[Cancel](#)

[Create database](#)

Create Crawler



Search

[Alt+S] Ask Amazon Q



United States (N. Virginia) 

Ashwin Raiyani (6091-2129-5001) 

lf-admin

AWS Glue > Crawlers > Add crawler

Notebooks

Job run monitoring

Data Catalog tables

Data connections

Workflows (orchestration)

Zero-ETL integrations [New](#)

▼ Data Catalog

Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

► Data Integration and ETL

► Legacy pages

What's New 

Documentation 

AWS Marketplace

☒ Enable compact mode

☒ Enable new navigation

Step 1

☒ Set crawler properties

Step 2

☐ Choose data sources and classifiers

Step 3

☐ Configure security settings

Step 4

☐ Set output and scheduling

Step 5

☐ Review and create

Set crawler properties

Crawler details [Info](#)

Name

Name can be up to 255 characters long. Some character set including control characters are prohibited.

Description - optional

Descriptions can be up to 2048 characters long.

► Tags - optional

Use tags to organize and identify your resources.

Cancel

Next

 CloudShell

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GlueCrawlerCustomerRole







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Global ▾

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☰

IAM > Roles > GlueCrawlerCustomerRole

Identity and Access Management (IAM)

Q Search IAM

Dashboard

▼ Access Management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Temporary delegation requests

New

▼ Access reports

Access Analyzer

Resource analysis New

Unused access

Summary

Edit

Creation date

February 26, 2026, 00:04 (UTC+05:30)

Last activity

✓ 12 hours ago

ARN

 arn:aws:iam::609121295001:role/GlueCrawlerCustomerRole

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (3) Info



Simulate 

Remove

Add permissions ▾

You can attach up to 10 managed policies.

Q Search

Filter by Type

All types ▾

< 1 > 

☐	Policy name 	Type	Attached entities
☐	  AmazonS3FullAccess	AWS managed	3
☐	  AWSGlueServiceRole	AWS managed	1
☐	  AWSLakeFormationDataAdmin	AWS managed	1

▶ Permissions boundary (not set)

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CSV → Crawler → Table Schema (Discover Schema)



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AWS Glue > Crawlers

Notebooks

Job run monitoring

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► Legacy pages

What's New [↗](#)

Documentation [↗](#)

AWS Marketplace

☒ Enable compact mode

☒ Enable new navigation

📘

Announcing new optimization features for Apache Iceberg tables

Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion. [Learn more](#) ↗

Crawlers

A crawler connects to a data store, progresses through a prioritized list of classifiers to determine the schema for your data, and then creates metadata tables in your data catalog.

Crawlers (2) [Info](#)

Last updated (UTC)
February 26, 2026 at 19:10:41

🔄

Action ▾

Run

Create crawler

< 1 >

⚙️

☐	Name ▾	State ▾	Schedule	Last run ▾	Last run times... ▾	Log	Table changes fr...
☐	CustomerCrawler	🟢 Ready		🟢 Succeeded	February 26, 2026...	View log ↗	1 created
☐	purchases-crawler	🟢 Ready		🟢 Succeeded	February 24, 2026...	View log ↗	1 created

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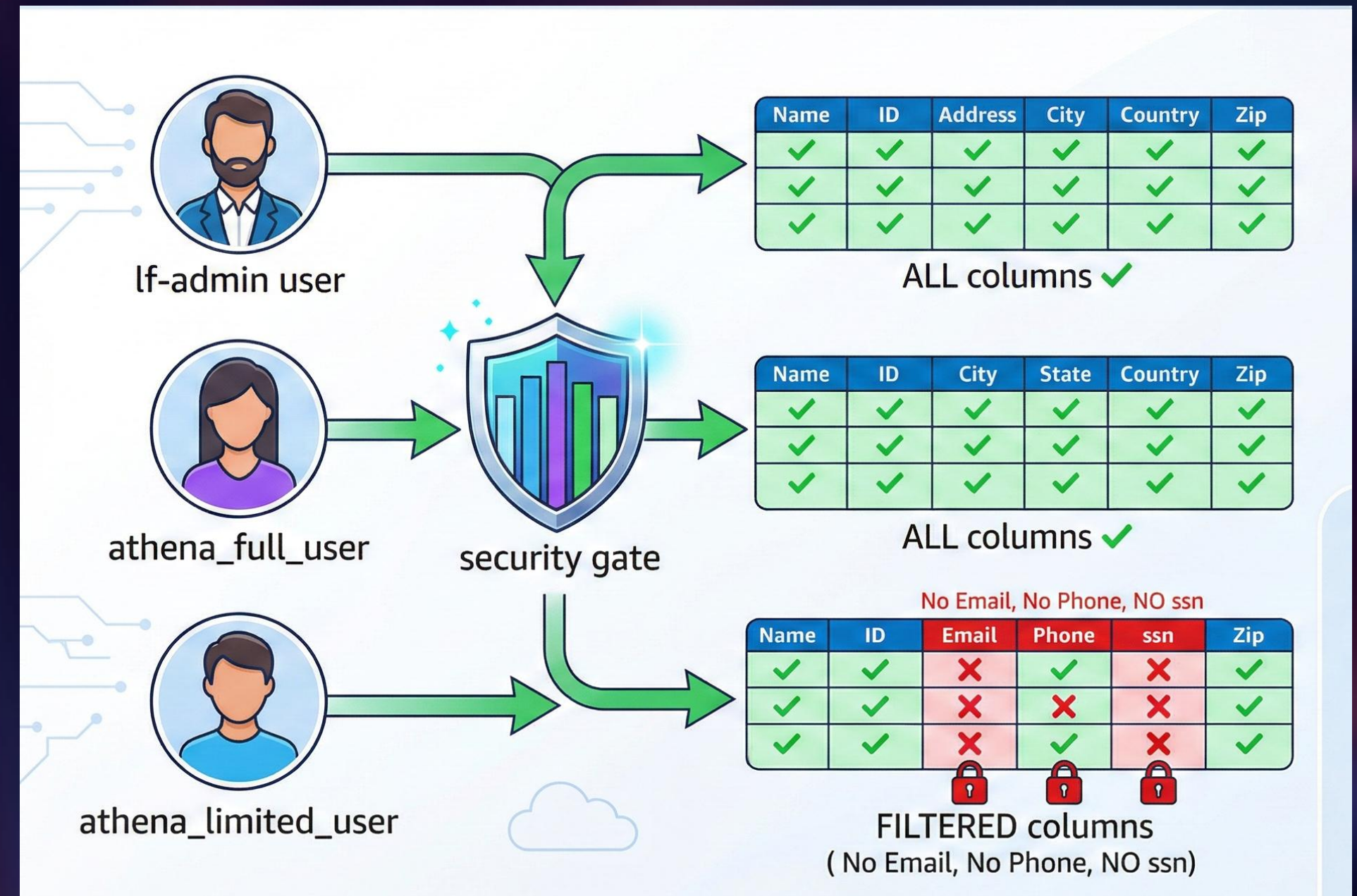
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#3. AWS Lake Formation – Security Guard

With Lake Formation:

- ✓ Grant permissions by column
- ✓ Automatic enforcement
- ✓ This happens at the DATA LAYER. Not the application layer. (Athena, QuickSight, Redshift Spectrum)



#3. AWS Lake Formation – Administration

aws





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AWS Lake Formation

> Dashboard

Databases

Tables and Materialized Views

Views

Data filters

Data sharing

Crawlers

▼ Permissions

Data permissions

LF-Tags and permissions

Hybrid access mode

Data locations

▼ Administration

Administrative roles and tasks

Data lake locations

Application integration settings

IAM Identity Center integration

Data Catalog settings

▼ Ingestion

Blueprints

▶ Data lake setup

Quickly set up your data lake in Lake Formation.

Recent access activity

Catalog optimization

Recent access activity (100/100) More available

 View event

Recent access activity for your data lake in AWS Cloudtrail. Events can take several minutes to appear in Cloudtrail and are limited to the last 90 days.

< 1 2 ... >



	Event name	Principal	Alert time
☐	BatchGrantPermissions	lf-admin	February 26, 2...
☐	ListDataCellsFilter	lf-admin	February 26, 2...
☐	ListLFTags	lf-admin	February 26, 2...
☐	ListDataCellsFilter	lf-admin	February 26, 2...
☐	DescribeLakeFormationIdentityCenterConfiguration	lf-admin	February 26, 2...
☐	ListLFTags	lf-admin	February 26, 2...
☐	ListLFTags	lf-admin	February 26, 2...

 CloudShell

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#3. AWS Lake Formation – Data Locations

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Search

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lf-admin

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AWS Lake Formation > Data locations

Databases

Tables and Materialized Views

Views

Data filters

Data sharing

Crawlers

▼ Permissions

Data permissions

LF-Tags and permissions

Hybrid access mode

Data locations

▼ Administration

Administrative roles and tasks

Data lake locations

Application integration settings

IAM Identity Center integration

Data Catalog settings

▼ Ingestion

Blueprints

Successfully added admins

Data locations (0)

Choose a storage location for which to review, grant or revoke user permissions.

S3 URI

Q e.g.: s3://bucket/prefix/

View

Browse

< 1 ... >

⚙

Principal	Principal type	Resource	Owner account ID	Grantable
⌛				

https://us-east-1.console.aws.amazon.com/lakeformation/home?region=us-east-1#location-...

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#3. AWS Lake Formation – Data Permissions



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If-admin

 AWS Lake Formation > Data permissions

AWS Lake Formation <

Dashboard

▼ Data Catalog

Catalogs [Updated](#)

Databases

Tables and Materialized Views

Views

Data filters

Data sharing

Crawlers [↗](#)

▼ Permissions

[Data permissions](#)

LF-Tags and permissions

Hybrid access mode

Data locations

▼ Administration

Administrative roles and tasks

Data lake locations

Application integration settings

IAM Identity Center integration

Too many permissions? Filter by database or table. In the navigation page, choose **Databases** or **Tables**. Then choose a database or table, and on the **Actions** menu, choose **View Permissions**.

Data permissions (12)

 [Revoke](#) [Grant](#)

☐

Principal ▲

Princip... ▼

Princip... ▼

Resour... ▼

Database ▼

Table ▼

Resource ▼

Catalog ▼

LF-Tag ex...

Perm

☐

AmazonS...

IAM role

arn:aws:ia...

Catalog

-

-

-

60912129...

-

All, A

☐

AmazonS...

IAM role

arn:aws:ia...

Database

sagemak...

-

sagemake...

60912129...

-

All, A

☐

AmazonS...

IAM role

arn:aws:ia...

Table

sagemak...

churn

churn

60912129...

-

All, A

☐

AmazonS...

IAM role

arn:aws:ia...

Column

sagemak...

churn

Included: ...

60912129...

-

Select

☐

athena_fu...

IAM user

arn:aws:ia...

Data locat...

-

-

s3://datal...

60912129...

-

Data

☐

athena_li...

IAM user

arn:aws:ia...

Data locat...

-

-

s3://datal...

60912129...

-

Data

☐

GlueCrawl...

IAM role

arn:aws:ia...

Database

customerdb

-

customerdb

60912129...

-

Creat

☐

GlueCrawl...

IAM role

arn:aws:ia...

Data locat...

-

-

s3://datal...

60912129...

-

Data

☐

GlueCrawl...

IAM role

arn:aws:ia...

Table

customerdb

customer...

customer...

60912129...

-

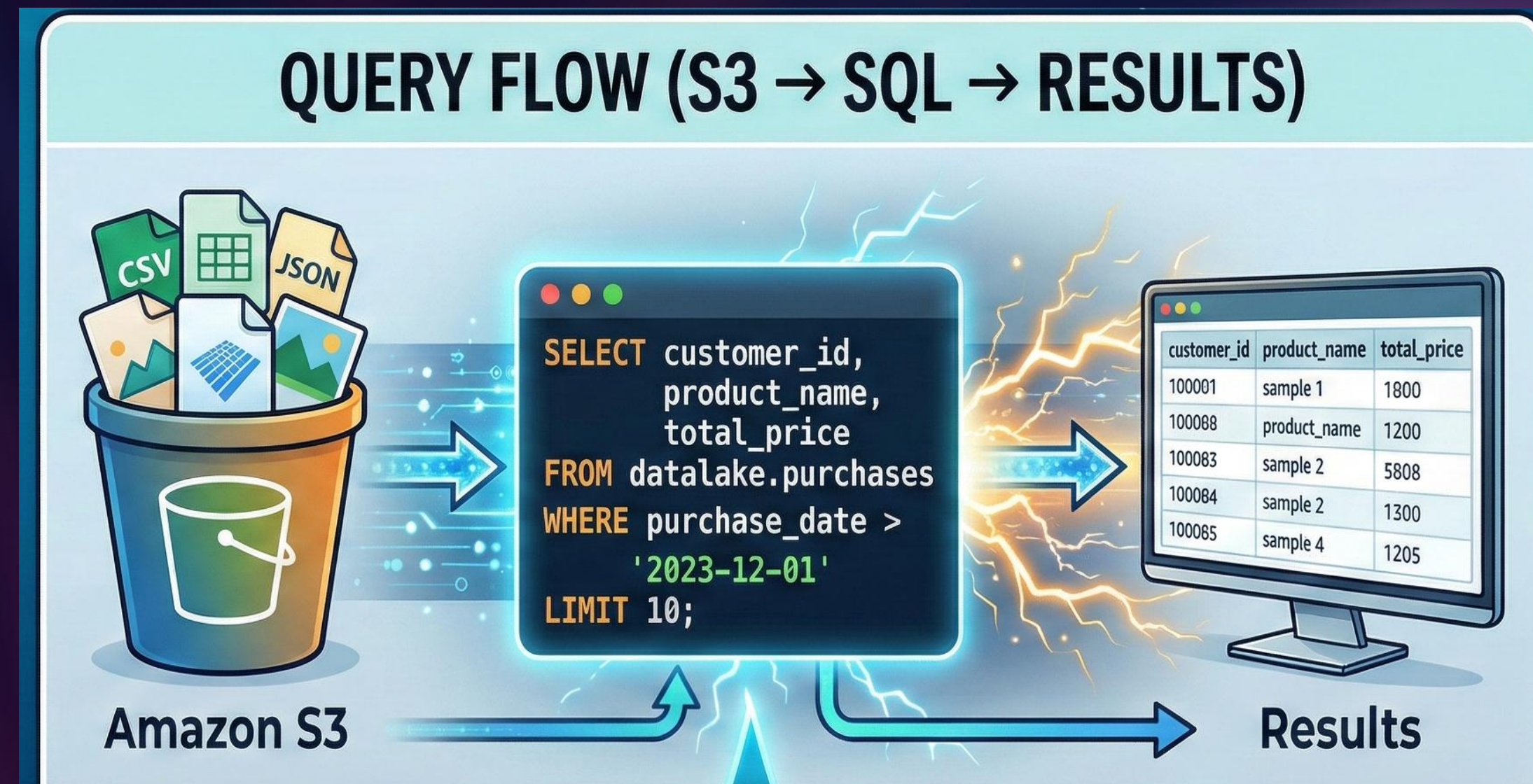
All, A

https://us-east-1.console.aws.amazon.com/lakeformation/home?region=us-east-1#permissi...

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#4. Amazon Athena - SQL Superpowers for S3

- Serverless SQL query engine
- Query S3 data directly (no loading!)
- Standard SQL (SELECT, JOIN, WHERE...)
- Pay per query (\$5 per TB scanned)





Permissions

Data Permissions

Resource Link Permissions

Governance

Setting Up Two Users:

2024-10-27 | Demo Environment

Grant Permissions

User Persona	User ID	Granted Permissions	Columns	Access Details
 Admin User (You)	admin@demo.internal	SELECT, DESCRIBE	ALL (9 columns)	Can see email, phone 
 Analyst User (Restricted)	analyst@demo.internal	SELECT only	INCLUDE specific (7 columns)	Cannot see email  , Cannot see phone 

 ADMIN VIEW

ID	Name	Date	Amount	Product	Region	Email 	Phone 	Dept
								

ADMIN VIEW
Full Access (9/9 Columns)

 ANALYST VIEW (Restricted)

Notices: PII Hiding Enabled

ID	Name	Date	Amount	Product	Region	Email 	Phone 	Dept
								

ANALYST VIEW (Restricted)
Filtered Access (7/9 Columns)



#4. AWS Athena – Full Access

[Provide feedback](#)

Multi-session disabled ▼

English ▼



IAM user sign in ⓘ

Account ID or alias [\(Don't have?\)](#)☐ Remember this account

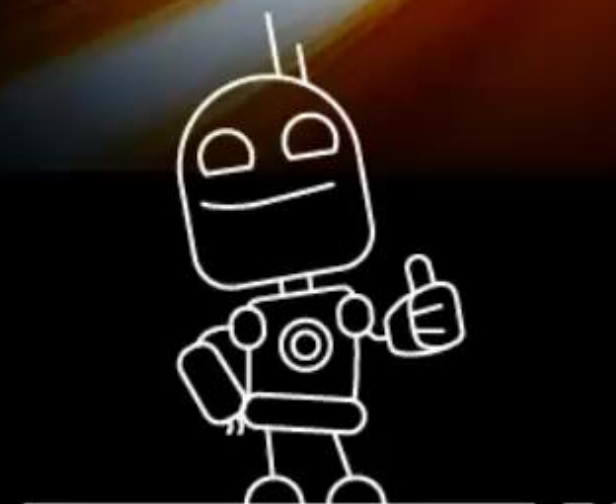
IAM username

Password

☐ Show Password[Having trouble?](#)[Sign in](#)[Sign in using root user email](#)[Create a new AWS account](#)

Amazon Lightsail

Lightsail is the easiest way
to get started on AWS

[Learn more »](#)

#4. AWS Athena – Limited Access



[Alt+S] 



United States (N. Virginia) ▼

Account ID: 6091-2129-5001 ▼
athena_limited_user

Amazon Athena > Query editor



Query in Amazon SageMaker Unified Studio

Use your IAM role to analyze and build with your existing AWS resources in a single data and AI development environment.

[Set up](#)

☒ Workgroup ☐ DataZone environment
primary ▼

Editor

Recent queries

Saved queries

Query settings

 Athena now supports typeahead code suggestions to speed up SQL query development
Typeahead suggestions are turned on by default. You can change this setting in query editor preferences.

Edit preferences

Data

Data source

AwsDataCatalog ▼

Catalog

None ▼

Database

customerdb ▼

Tables and views

Create ▼

Query 6

1

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 **Stay Connected:**



Ashwin Raiyani

 [in/ashwinraiyni/](https://www.linkedin.com/in/ashwinraiyni/)

 **Questions?**

“No question is too basic – we all started somewhere!”



Thanks for listening

